

# Vital Signs Driven Early Warning System using Survival Models

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**Abstract**—Healthcare systems operating in rural and resource-limited settings often rely on basic vital parameters to evaluate patient health due to restricted access to advanced diagnostic technologies such as CT scans, MRI, and laboratory testing. This study introduces a cost-effective and interpretable early warning system based on time-to-event survival analysis to estimate both the likelihood and timing of patient deterioration. The proposed model utilizes commonly available physiological indicators, including heart rate, blood pressure, oxygen saturation, respiratory rate, and body temperature, along with demographic attributes such as age and comorbid conditions. In addition, a red-flag alert mechanism is integrated to identify abnormal long-term patterns, such as sustained oxygen decline, persistent fever, or delayed recovery, and to recommend timely medical intervention. The Cox Proportional Hazards-based survival model achieved a Concordance Index (C-index) of 0.675, indicating moderate performance in ranking patients according to time-dependent risk. The results demonstrate that the system provides time-aware risk predictions while remaining computationally efficient and interpretable. Overall, the proposed approach supports early detection, assists community healthcare workers, and improves healthcare delivery in underserved populations.

**Index Terms**—Survival Analysis, Cox Proportional Hazards Model, Early Warning Systems, Red Flag Layer, Time-to-Event Prediction, Vital Signs Monitoring, Health Risk Prediction, Interpretable Machine Learning.

## I. INTRODUCTION

Healthcare systems in rural and resource-constrained regions encounter significant difficulties due to inadequate access to advanced medical infrastructure and diagnostic technologies. In many underserved communities, essential tools such as CT and MRI scanners are often unavailable, which limits the ability to perform timely diagnoses and informed clinical decision-making. Although artificial intelligence has demonstrated considerable promise in enhancing healthcare services and enabling predictive analytics [1], [21], the practical implementation of such advanced AI-driven solutions in low-resource environments continues to be a major challenge.

In such settings, medical practitioners largely depend on fundamental physiological indicators—such as heart rate, blood pressure, oxygen saturation (SpO<sub>2</sub>), respiratory rate, and body temperature—to evaluate a patient’s condition. While these measurements are cost-effective and readily obtainable, deriving meaningful insights from them for the early identification of health deterioration remains a complex task. Clinical prediction models have traditionally been employed to assess patient risk [25]; however, many conventional methods are limited in their ability to account for the temporal progression of a patient’s health status.

Survival analysis offers an effective approach for modeling time-to-event data in healthcare applications. Among the available techniques, the Cox Proportional Hazards model is extensively utilized due to its simplicity and interpretability in clinical research settings [13]. In recent years, advanced machine learning-based survival methods, including Random Survival Forests and deep survival models, have shown enhanced predictive capabilities [15], [16]. However, these approaches often involve increased complexity, which can reduce their transparency and interpretability in practical medical use.

Interpretability plays a crucial role in healthcare applications, as transparent and explainable decision-making is essential for clinical practice [2], [23]. Many contemporary machine learning approaches function as black-box models, limiting their usability in real-world scenarios, particularly for frontline healthcare workers operating in resource-constrained settings.

To overcome these challenges, this study introduces a cost-efficient and interpretable early warning framework based on time-to-event survival analysis. The proposed system leverages readily available data, including vital signs, patient age, and comorbid conditions, to predict both the probability and timing of potential health deterioration. Furthermore, it integrates a red-flag alert mechanism to identify abnormal physiological patterns over time, enabling timely intervention. By combining temporal risk prediction with actionable alerts, the framework is designed to facilitate early detection, support community healthcare providers, and enhance the overall quality of healthcare delivery in underserved regions.

## II. RELATED WORK

Survival analysis has been extensively utilized in healthcare to model time-to-event outcomes, including mortality risk, disease progression, and duration of hospital stay. Huang et al. [1] developed a temporal similarity-driven approach for estimating hospital length of stay by incorporating time-varying clinical treatment patterns, demonstrating that the inclusion of longitudinal information enhances predictive accuracy compared to static models. In a related study, Jing et al. [2] proposed a deep learning-based survival analysis framework using ranking techniques, which improved both risk stratification and prediction performance by effectively modeling complex nonlinear interactions among clinical features.

Extensive benchmarking studies have been carried out to assess the robustness of various survival prediction method-

ologies. Herrmann et al. [4] conducted a detailed evaluation of multiple survival models using multi-omics datasets, revealing that model performance can vary considerably depending on data properties and the chosen modeling approach. In a more recent study, Baniecki et al. [8] introduced multi-modal survival models that combine radiological imaging data with textual clinical reports for predicting hospital length of stay, demonstrating that the integration of structured and unstructured data sources can significantly enhance survival prediction accuracy [38].

Interpretability has become an essential aspect of clinical survival modeling, ensuring that predictive outcomes can be clearly understood and trusted by healthcare professionals. Stiglic et al. [3] presented a systematic review of interpretability methods in healthcare machine learning, classifying them into model-specific and model-agnostic categories. Wang et al. [5] introduced a counterfactual explanation approach for survival prediction in cardiovascular ICU settings, allowing clinicians to evaluate how changes in input features can impact predicted outcomes. Additionally, Krzyżiński et al. [9] proposed SurvSHAP(t), a time-dependent attribution technique that captures how the importance of predictors varies over the survival horizon. Ter-Minassian et al. [10] further enhanced interpretability by introducing the Median-SHAP method, which improves the consistency and reliability of SHAP-based explanations in survival analysis.

Beyond research focused specifically on survival model interpretability, several studies have explored the broader application of artificial intelligence in healthcare systems. Rajpurkar et al. [6] examined the role of AI in enhancing diagnostic accuracy, improving workflow efficiency, and supporting clinical decision-making, while also addressing key challenges such as bias and limitations in data quality. Similarly, Ooge et al. [7] analyzed visual analytics approaches for interpreting AI models in healthcare, highlighting the critical importance of transparency and fostering user trust in real-world clinical environments [40].

Although existing research has achieved notable advancements in both survival prediction performance and model interpretability, the majority of these approaches are designed for high-resource environments, rely on multi-omics data, or involve computationally demanding deep learning techniques. Comparatively little attention has been given to the development of lightweight and interpretable survival-based early warning systems specifically suited for resource-limited settings, where only basic physiological data are available. Consequently, there is a clear need for an efficient time-to-event risk modeling framework that combines interpretable survival analysis with practical alert mechanisms, making it more applicable to low-resource healthcare environments.

### III. OBJECTIVES

The main aim of this study is to design a computationally efficient and cost-effective early warning system for predicting patient health deterioration using routinely available vital signs and demographic information. The proposed approach utilizes

time-to-event survival analysis methods to estimate both the likelihood and timing of deterioration in a clinically relevant way. Survival-based modeling techniques have proven to be effective for capturing temporal clinical outcomes, especially in applications such as hospital length-of-stay prediction and risk assessment, where incorporating time-dependent information enhances predictive accuracy [1].

Another important objective of this work is to maintain interpretability within the survival modeling framework, ensuring that the generated risk predictions are transparent and easily understood in a clinical context. Interpretability has become a fundamental requirement in healthcare AI applications, particularly when model outputs are used to support clinical decision-making. Previous studies have highlighted the need for explainable survival analysis techniques that offer clear insights into feature contributions and hazard relationships for medical practitioners [3]. In line with this, the present study employs an interpretable Cox Proportional Hazards model to achieve a balance between transparency and predictive performance.

Finally, this study seeks to assess the performance of the proposed system using established survival analysis evaluation metrics, particularly discrimination capability measured by the concordance index. Recent large-scale benchmarking efforts in survival prediction have emphasized the necessity of rigorous evaluation and comparative analysis across different models to ensure reliability and generalizability in clinical applications [4]. By adhering to these standardized evaluation approaches, this work examines the practicality and effectiveness of the proposed early-warning system for implementation in resource-constrained healthcare settings.

## IV. DATA AND PROCESSING

### A. Dataset

The dataset used in this study is obtained from the MIMIC-IV Clinical Database (Demo Version 2.2), a publicly accessible and de-identified healthcare dataset that contains detailed clinical information from intensive care unit (ICU) patients. The demo version comprises records of 100 individual patients and is commonly utilized for academic research, model validation, and the evaluation of methodologies in clinical predictive analytics.

The dataset includes 275 hospital admissions and 140 ICU stays, with a total of 668,862 time-stamped clinical observations. These longitudinal data records reflect the temporal evolution of physiological variables and patient health status, making them suitable for time-to-event survival analysis aimed at predicting patient deterioration.

The dataset includes regularly recorded physiological parameters such as heart rate, blood pressure, oxygen saturation ( $SpO_2$ ), respiratory rate, and body temperature, which are critical for continuous patient monitoring and form the core predictors in early warning systems. Alongside these clinical measurements, demographic factors including patient age and recorded comorbid conditions are also incorporated to enable more personalized and accurate risk assessment.

The structure of the dataset supports time-to-event analysis by incorporating longitudinal, time-stamped observations, allowing for the identification of both deterioration events and censored instances where no event is observed within the study period. With several hundred admissions and more than six hundred thousand clinical records, the dataset offers adequate temporal richness and variability for evaluating the performance of the Cox Proportional Hazards-based early warning system.

### B. Data Pre-Processing

The raw clinical data obtained from the MIMIC-IV database includes missing entries, inconsistencies in measurements, and irregular intervals in time-stamped observations. As a result, a structured preprocessing pipeline was implemented to improve data quality and prepare the dataset for effective survival modeling.

Initially, relevant physiological variables were extracted and screened using clinically valid ranges to eliminate unrealistic outliers. Missing data were handled through forward filling to maintain continuity in time-series observations, along with mean imputation where suitable. Furthermore, the data were resampled into uniform hourly intervals to achieve temporal consistency across all patient records.

For the purpose of survival analysis, deterioration events were identified based on clinically meaningful abnormal trends observed in physiological parameters. Instances corresponding to such deterioration were labeled as event occurrences (Event = 1), while cases in which no deterioration was observed within the study period were treated as censored observations (Event = 0). The processed data were subsequently organized into a time-to-event structure compatible with Cox Proportional Hazards modeling.

TABLE I  
DATA PRE-PROCESSING RULES

| Parameter                             | Preprocessing Rule              |
|---------------------------------------|---------------------------------|
| Heart Rate (HR)                       | 40–180 bpm (outliers removed)   |
| Blood Pressure (BP)                   | 70–200 mmHg                     |
| Oxygen Saturation (SpO <sub>2</sub> ) | 85–100%                         |
| Respiratory Rate (RR)                 | 8–40 breaths/min                |
| Temperature                           | 35–40°C                         |
| Missing Values                        | Forward fill / Mean imputation  |
| Time Interval                         | Aligned to hourly intervals     |
| Event Label                           | 1 = Deterioration, 0 = Censored |

### C. Normalization and Scaling

Normalization and feature scaling techniques were applied to maintain numerical consistency among physiological variables recorded on different scales. Key vital parameters, including heart rate, blood pressure, oxygen saturation, respiratory rate, and body temperature, were standardized to a common range to avoid the dominance of variables with larger magnitudes during the model estimation process.

Feature scaling plays a crucial role in improving numerical stability during the optimization of the Cox Proportional Hazards model and helps achieve balanced estimation of hazard

coefficients across different predictors. This process supports better convergence during model training and contributes to more consistent and reliable risk predictions within the survival analysis framework.

## V. METHODOLOGY

The Cox Proportional Hazards (Cox PH) model is a commonly used statistical method in survival analysis for handling time-to-event data. Such survival modeling approaches have been widely applied in healthcare-related prediction tasks, including estimating hospital length of stay and performing clinical risk stratification [2], [4], [8]. In contrast to traditional classification techniques that focus solely on predicting whether an event will occur, the Cox model evaluates the hazard function, allowing for the assessment of risk over time.

The Cox Proportional Hazards model is a semi-parametric approach that expresses the hazard of an event as a function of individual-specific covariates. For a patient characterized by the covariate vector  $X = (x_1, x_2, \dots, x_p)$ , the hazard function can be written as

$$h(t|X) = h_0(t) \exp(\beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p) \quad (1)$$

where  $h_0(t)$  denotes the baseline hazard, and  $\beta_i$  corresponds to the regression coefficient associated with the  $i^{th}$  covariate. This representation allows the model to quantify the relative impact of clinical features on risk while maintaining flexibility in capturing the underlying time-varying baseline hazard.

A fundamental assumption of the Cox model is the proportional hazards property, which implies that the hazard ratios between individuals remain constant over time. The model coefficients are interpreted using their exponential form,  $\exp(\beta_i)$ , representing the hazard ratio associated with each covariate. A hazard ratio greater than one indicates an elevated risk, while a value less than one reflects a reduced risk. Recent advancements in survival analysis and explainable modeling have highlighted the significance of interpretable hazard estimation for supporting clinical decision-making systems[9], [10].

Owing to its ability to effectively manage right-censored data and produce interpretable risk estimates, the Cox Proportional Hazards model continues to serve as a reliable baseline method for survival prediction tasks in healthcare and the development of early warning systems [4], [8].

### A. Architecture Overview

The overall design of the proposed survival-based early warning framework is depicted in Fig. 1. The system is structured to estimate the time-dependent risk of patient health deterioration by utilizing structured clinical covariates within a Cox Proportional Hazards modeling approach.

1) *Covariates*: The framework begins with a set of covariates  $X = (X_1, X_2, \dots, X_n)$ , representing patient-specific clinical features. These variables include commonly monitored physiological measurements such as heart rate, blood pressure, oxygen saturation, respiratory rate, and body temperature,

along with demographic information and indicators of comorbid conditions. Each covariate acts as an individual predictor that contributes to the overall hazard of patient deterioration. The use of structured clinical variables in survival prediction has been extensively applied in healthcare risk modeling studies [4], [8].

2) *Input Layer*: The input layer accepts structured covariate values in the form of numerical vectors, ensuring a standardized representation of features for subsequent processing. While this layer does not directly perform prediction, it organizes patient-specific measurements into a consistent format that is appropriate for survival analysis and hazard estimation.

3) *Hidden Layers and Feature Transformation*: The intermediate processing layers model interactions among clinical variables and derive latent feature representations. These transformations facilitate the capture of complex relationships between physiological parameters that may not exhibit linear separability. Recent studies in survival modeling have highlighted the importance of structured feature learning to enhance time-to-event prediction performance [4].

4) *Dropout Regularization*: To improve generalization and mitigate overfitting, dropout regularization is applied during the training phase. In this approach, a fraction of neurons is randomly deactivated in each iteration, reducing dependency on specific features and enhancing model robustness across diverse patient groups. Such regularization techniques are especially valuable in healthcare survival modeling, where datasets are often limited in size [8].

5) *Regression Layer: Cox Proportional Hazards Formulation*: The regression layer applies the Cox Proportional Hazards formulation to estimate the risk of an event. For an individual characterized by the covariate vector  $X$ , the hazard function is expressed as

$$h(t|X) = h_0(t) \exp(\beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n) \quad (2)$$

where  $h_0(t)$  represents the baseline hazard, and  $\beta_i$  denotes the learned coefficient associated with the  $i^{th}$  covariate. This semi-parametric formulation allows the model to estimate relative risk without requiring explicit specification of the baseline hazard function. Owing to its interpretability and capability to handle censored observations, the Cox model continues to be widely used in clinical survival prediction tasks [4], [9].

6) *Hazard Function Estimation*: The estimated hazard function reflects the instantaneous risk of patient deterioration at a given time  $t$ , conditioned on the associated covariates. In contrast to binary classification models, this formulation enables continuous time-dependent risk assessment, thereby supporting dynamic monitoring of patient health status.

7) *Hazard Ratio Output*: The final output of the architecture is the hazard ratio, derived from the exponential transformation of the regression coefficients. These ratios measure the relative change in risk associated with each predictor, where values greater than one indicate increased risk and values less than one correspond to decreased risk. Such interpretable risk quantification is essential for clinical decision support and effective risk stratification frameworks [9], [10].

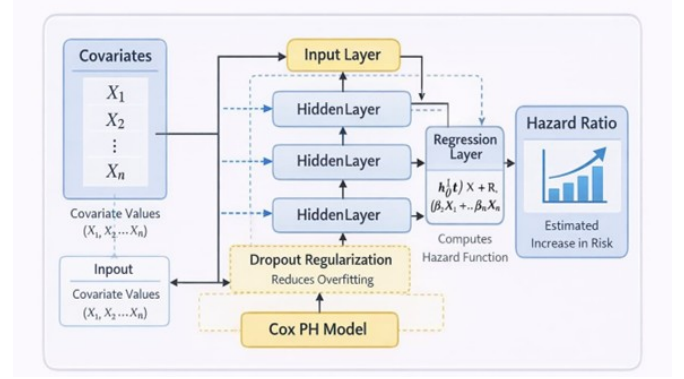


Fig. 1. Proposed Cox-based Survival Prediction Architecture.

Fig. 1 illustrates the overall architecture of the Cox-based survival prediction framework. The model starts with structured patient covariates  $(X_1, X_2, \dots, X_n)$ , which represent physiological measurements and clinical characteristics. These inputs are processed through an input layer that arranges the feature vector for subsequent computation. Intermediate transformation layers capture interactions among the covariates and refine feature representations before performing regression-based hazard estimation.

A dropout-based regularization technique is applied during training to mitigate overfitting and enhance the model's generalization capability. The regression layer follows the Cox Proportional Hazards formulation, where a weighted linear combination of covariates is exponentiated and scaled by the baseline hazard function to estimate the time-dependent hazard.

The final output of the architecture is the hazard ratio, which represents the relative risk of deterioration based on patient-specific covariates. This formulation supports interpretable survival prediction while preserving flexibility in modeling time-to-event outcomes.

## B. Training and Testing

The Cox Proportional Hazards model is trained on time-to-event clinical data, which includes survival times and event indicators corresponding to patient deterioration or censored observations. The dataset is partitioned into training and testing subsets to facilitate proper model validation. During training, the model estimates regression coefficients for each covariate by maximizing the partial likelihood function, thereby learning the relationship between physiological features and patient-specific risk.

After training, the estimated coefficients are applied to unseen test data to assess predictive performance. The model calculates hazard ratios for each patient, reflecting their relative risk of health deterioration over time. This evaluation step ensures that the model generalizes effectively beyond the training dataset.

Model performance is evaluated using standard survival analysis metrics, including the Concordance Index (C-index) to assess ranking accuracy and calibration analysis to examine

the agreement between predicted and observed outcomes. This training and evaluation procedure ensures reliable, interpretable, and time-aware risk predictions.

### C. Evaluation Metrics

The Concordance Index (C-Index) is a widely adopted metric for evaluating survival analysis models. It quantifies the model’s ability to correctly rank patients according to their predicted risk of experiencing an event. Unlike conventional accuracy metrics that measure classification correctness, the C-Index assesses how well the model predicts the ordering of survival times.

The mathematical formulation of the C-Index is given by:

$$C = \frac{N_{conc}}{N_{comp}} \quad (3)$$

where: Concordant Pairs: Pairs of patients in which the patient who experiences the event earlier has a higher predicted risk score.

Comparable Pairs: Pairs where one patient experiences the event before the other and the earlier event is not censored.

The C-Index ranges from 0 to 1, with a value of 0.5 indicating random ranking and values approaching 1 representing better risk discrimination. This metric is particularly appropriate for survival models such as the Cox Proportional Hazards model, as it accounts for censored observations and evaluates the model’s ability to rank time-to-event outcomes effectively.

## VI. RESULTS

This section presents the experimental evaluation of the proposed Cox Proportional Hazards-based early warning framework using the MIMIC-IV demo dataset. The performance assessment emphasizes covariate effect estimation, patient risk stratification, and the analysis of red-flag alert distributions.

The Cox Proportional Hazards model facilitates time-dependent risk estimation while properly handling right-censored observations. Model performance was assessed using the Concordance Index (C-Index), which evaluates the agreement between predicted and observed event rankings. The proposed framework achieved a C-Index of 0.675, outperforming the baseline Cox model, which attained a C-Index of 0.645, demonstrating enhanced discriminative capability.

Fig. 2 depicts the Kaplan–Meier survival curves stratified according to predicted risk groups. The high-risk group shows a more pronounced decline in survival probability relative to the medium- and low-risk groups, demonstrating the Cox model’s effectiveness in discriminating patient risk and capturing time-dependent differences in survival outcomes. Risk stratification using the predicted hazard scores resulted in distinctly separated survival curves for the low-, medium-, and high-risk groups. This clear separation validates the model’s capability to differentiate patients according to their risk of health deterioration.

Fig. 3 presents the estimated log-hazard ratios along with their 95% confidence intervals for the selected covariates. Covariates with positive coefficients correspond to an increased

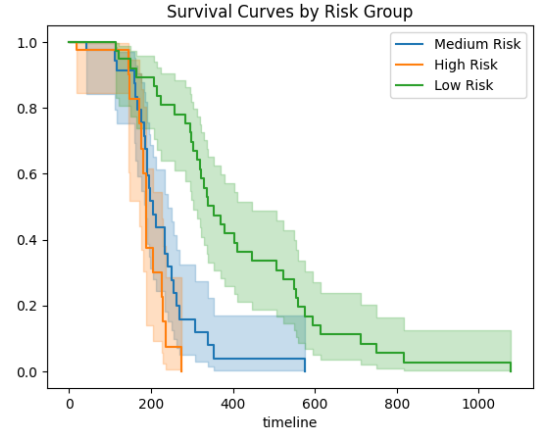


Fig. 2. Kaplan–Meier survival curves by predicted risk group.

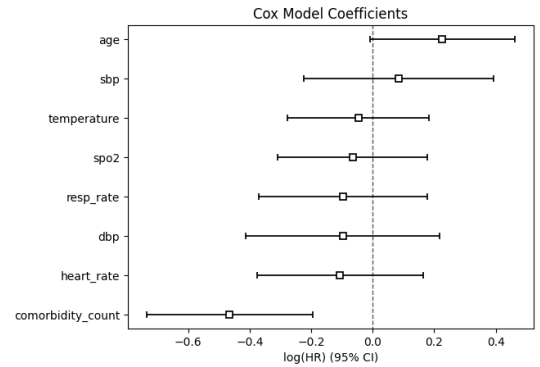


Fig. 3. Cox model log-hazard ratios with 95% confidence intervals.

risk of deterioration, whereas negative coefficients indicate a reduced risk. The confidence intervals offer insights into the stability and relative importance of each predictor.

Beyond survival prediction, the red-flag alert mechanism generates interpretable clinical indicators based on abnormal physiological patterns. The distribution of these alerts identifies the most prominent deterioration triggers and facilitates the early detection of high-risk patients.

Fig. 4 illustrates the distribution of red-flag alerts across physiological parameters. Although the majority of measurements fall within normal ranges, abnormal alerts—such as persistent fever, hypotension, and respiratory distress—highlight clinically significant deterioration patterns detected by the system.

Overall, the results indicate that the proposed framework successfully combines survival modeling with interpretable clinical alert generation, supporting effective monitoring of ICU patients.

## VII. CONCLUSION AND FUTURE WORK

This study introduced a structured survival modeling framework for analyzing time-to-event clinical data using the Cox Proportional Hazards (Cox PH) model. Survival modeling approaches have been extensively employed in healthcare for risk



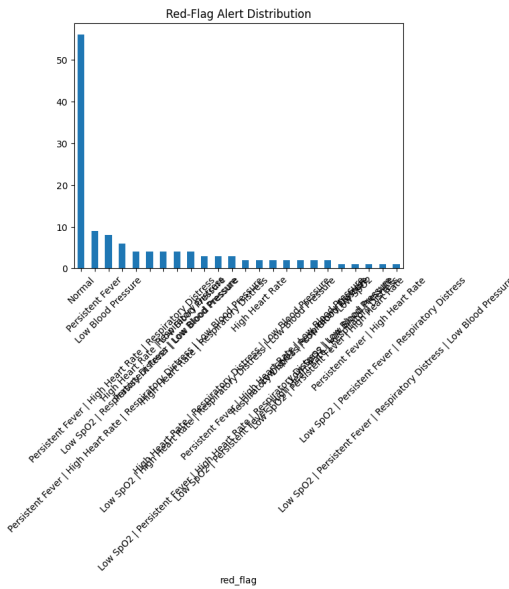


Fig. 4. Distribution of red-flag alerts across physiological conditions.

prediction and hospital outcome assessment [2], [4],[8]. By leveraging routinely collected physiological and demographic features, the proposed system allows for the estimation of hazard functions and survival probabilities in a time-dependent manner. In contrast to traditional classification-based early warning systems, the survival-based approach provides continuous risk estimates while appropriately accounting for censored observations.

The integration of a red-flag alert mechanism with survival probability estimation enhances continuous patient monitoring. Rather than depending solely on static threshold-based triggers, the framework combines statistical hazard modeling with longitudinal detection of abnormal physiological patterns. Experimental evaluation demonstrates improved discriminative capability, with the proposed model achieving a C-Index of 0.675 compared to 0.645 for the baseline Cox model. Additionally, the model provides interpretable hazard ratios, supporting transparent clinical risk assessment. The critical role of interpretability in survival prediction and healthcare decision support has been highlighted in recent studies [9], [10].

*Future Scope:* Although the proposed framework establishes a robust semi-parametric survival modeling approach, several extensions could further enhance its analytical capabilities. Incorporating time-varying covariates may improve hazard estimation across longitudinal measurements. Comparative analyses with alternative survival methods, such as Random Survival Forests and deep survival learning models, could provide comprehensive benchmarking across clinical scenarios [2], [4]. Furthermore, integrating advanced explainability techniques can strengthen transparency in time-dependent risk assessment [9]. Large-scale multi-center validation and implementation within electronic health record (EHR) systems would enhance generalizability and support practical deployment in clinical

settings.

Beyond technical performance, the proposed framework carries important societal implications. Early identification of high-risk patients can facilitate timely interventions, reduce preventable complications, and optimize the use of healthcare resources. By combining interpretable survival modeling with structured alert mechanisms, the system contributes to improved patient safety and supports more data-driven healthcare delivery.

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